

Credit Card Approval Prediction System

Project Description

The Credit Card Approval Prediction System is a machine learning based web application that predicts whether a credit card application will be approved or rejected using applicant details such as age, income, employment status, credit history and loan information.

Technical Architecture

User Input → Flask Application → Data Preprocessing → Machine Learning Model → Prediction Engine → Approval/Rejection Result.

Milestone 1: Dataset Preparation and Model Selection

Load the synthetic creditcard.csv dataset, preprocess the data, encode categorical variables and train multiple classification algorithms.

Milestone 2: Model Development

Train Logistic Regression, Decision Tree and Random Forest models and save the best model as model.pkl and label_encoders.pkl.

Milestone 3: Flask Backend Development

Develop app.py to load the trained model and provide prediction functionality through Flask routes.

Milestone 4: Frontend Development

Create attractive HTML and CSS pages to collect applicant information and display approval or rejection results.

Milestone 5: Deployment

Install requirements, run the Flask application locally and access the project through a web browser.

Conclusion

The project demonstrates the application of machine learning in banking and financial services by automating credit approval decisions.